

第四章 能带论

金属电导：

1. Ohm 定律: $U = IR$, $J = \sigma E$

2. Drude 模型:

$$m v_d = -e E \tau$$

$$J = -nev_d = \frac{ne^2 E}{m}$$

符合经验定律 $\frac{\sigma}{T} = \left(\frac{3}{2}\right)^2 \frac{k_B}{e}$ $\sigma = \frac{1}{2} v_d^2 \tau n e$, $\sigma = \frac{3}{2} n k_B$

不能说明: $\rho \sim T$ 关系, 电子自由程太短, 电子自由程对比热容贡献

能带论近似:

① 绝热近似: 电子实固定

② 单电子近似

③ 电子处于周期势场中

$$\begin{cases} \left[-\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r}) \right] \psi(\vec{r}) = E \psi(\vec{r}) \\ V(\vec{r} + \vec{R}_i) = V(\vec{r}), \quad \vec{R}_i = \sum_j n_j \vec{a}_j \end{cases}$$

1. 平移方程

$$\hat{T}(\vec{a}_i) \psi(\vec{r}) = \psi(\vec{r} + \vec{a}_i)$$

$$\hat{T}(\vec{a}_i) \hat{H} \psi(\vec{r}) = \hat{H} \psi(\vec{r} + \vec{a}_i)$$

满足 $[\hat{T}(\vec{a}_i), \hat{T}(\vec{a}_j)] = 0$

由于 $V(\vec{r})$ 有周期性:

$$\begin{aligned} \hat{T}(\vec{a}_i) \hat{H} \psi(\vec{r}) &= \left[-\frac{\hbar^2}{2m} \nabla_{\vec{r}+\vec{a}_i}^2 + V(\vec{r} + \vec{a}_i) \right] \psi(\vec{r} + \vec{a}_i) \\ &= \left[-\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r}) \right] \psi(\vec{r} + \vec{a}_i) = \hat{H} \hat{T}(\vec{a}_i) \psi(\vec{r}) \end{aligned}$$

$$\rightarrow [\hat{T}(\vec{a}_i), \hat{H}] = 0$$

2. 平移算符的本征值.

$$\textcircled{1} \quad \begin{cases} \hat{H} \varphi(\vec{r}) = E_n \varphi(\vec{r}) \\ T(\vec{a}_i) \varphi(\vec{r}) = \lambda(\vec{a}_i) \varphi(\vec{r}) \end{cases}$$

$$\hat{H} [T(\vec{a}_i) \varphi(\vec{r})] = T(\vec{a}_i) \hat{H} \varphi(\vec{r}) = E_n T(\vec{a}_i) \varphi(\vec{r})$$

$\Rightarrow \varphi(\vec{r}), T(\vec{a}_i) \varphi(\vec{r})$ 都是 $\hat{H}$ 的本征函数, 本征值为 E_n

$$|T(\vec{a}_i) \varphi(\vec{r})|^2 = |\lambda(\vec{a}_i) \varphi(\vec{r})|^2 = |\varphi(\vec{r})|^2$$

$$|\lambda(\vec{a}_i)|^2 = 1$$

$$\textcircled{2} \quad T(\vec{a}_i) T(\vec{a}_j) \varphi(\vec{r}) = \lambda(\vec{a}_i) \lambda(\vec{a}_j) \varphi(\vec{r}) = \varphi(\vec{r} + \vec{a}_i + \vec{a}_j)$$

$$T(\vec{a}_i + \vec{a}_j) \varphi(\vec{r}) = \lambda(\vec{a}_i + \vec{a}_j) \varphi(\vec{r}) = \varphi(\vec{r} + \vec{a}_i + \vec{a}_j)$$

$$\Rightarrow \lambda(\vec{a}_i) \lambda(\vec{a}_j) = \lambda(\vec{a}_i + \vec{a}_j)$$

同时满足①、②只有 $\lambda(\vec{a}) = e^{i\vec{k} \cdot \vec{a}}$

由一维边界条件,

$$\varphi(r_i + N_i \vec{a}_i) = \lambda^{N_i}(\vec{a}_i) \varphi(\vec{r}) = \varphi(\vec{r})$$

$$\lambda^{N_i}(\vec{a}_i) = e^{i N_i \vec{k} \cdot \vec{a}_i} = 1$$

$$\Rightarrow N_i \vec{k} \cdot \vec{a}_i = 2\pi N_i$$

则将 $\vec{k}$ 写成 $\vec{k} = \sum_i \frac{h_i}{N_i} \vec{b}_i$, $\vec{b}_i$ 为倒格矢

二、Bloch定理和Bloch波.

Bloch定理: 单电子波函数 $\psi_{\vec{k}}^n(\vec{r})$

$$\begin{aligned}\psi_{\vec{k}}^n(\vec{r} + \vec{R}_L) &= T(\vec{R}_L) \psi_{\vec{k}}^n(\vec{r}) \quad \vec{R}_L = \sum_i L_i \vec{a}_i \\ &= \prod_i T^L(\vec{a}_i) \cdot \psi_{\vec{k}}^n(\vec{r}) \\ &= e^{i\vec{k} \cdot \vec{R}_L} \psi_{\vec{k}}^n(\vec{r})\end{aligned}$$

$$\Rightarrow \psi_{\vec{k}}^n(\vec{r}) = e^{i\vec{k} \cdot \vec{r}} \cdot u_{\vec{k}}^n(\vec{r})$$

其中 $u_{\vec{k}}^n(\vec{r})$ 是调制因子, $u_{\vec{k}}^n(\vec{r} + \vec{R}_L) = u_{\vec{k}}^n(\vec{r})$ 周期性.

$$\begin{aligned}\psi_{\vec{k}}^n(\vec{r} + \vec{R}_L) &= e^{i\vec{k} \cdot (\vec{r} + \vec{R}_L)} u_{\vec{k}}^n(\vec{r} + \vec{R}_L) \\ &= e^{i\vec{k} \cdot \vec{R}_L} \cdot e^{i\vec{k} \cdot \vec{r}} u_{\vec{k}}^n(\vec{r}) \\ &= e^{i\vec{k} \cdot \vec{R}_L} \psi_{\vec{k}}^n(\vec{r})\end{aligned}$$

Bloch波: $u_{\vec{k}}^n(\vec{r})$

三、Bloch方程特征

1. 对一个确定的 $\vec{k}$, 有无穷多个特征值 $E_n(\vec{k})$ 和特征函数 $\psi_n(\vec{r})$

$$\hat{H} e^{i\vec{k} \cdot \vec{r}} u_{\vec{k}}^n(\vec{r}) = E_n(\vec{k}) e^{i\vec{k} \cdot \vec{r}} u_{\vec{k}}^n(\vec{r})$$

$$\hat{H} u_{\vec{k}}^n(\vec{r}) = E_n(\vec{k}) u_{\vec{k}}^n(\vec{r})$$

$$\text{其中 } \hat{H} = e^{-i\vec{k} \cdot \vec{r}} \hat{H} e^{i\vec{k} \cdot \vec{r}}$$

$$= e^{-i\vec{k} \cdot \vec{r}} \left[-\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r}) \right] e^{i\vec{k} \cdot \vec{r}}$$

$$= e^{-i\vec{k} \cdot \vec{r}} \left[-\frac{\hbar^2}{2m} \nabla \cdot (i\vec{k} e^{i\vec{k} \cdot \vec{r}}) + V(\vec{r}) e^{i\vec{k} \cdot \vec{r}} \right]$$

$$= e^{-i\vec{k} \cdot \vec{r}} \left[-\frac{\hbar^2}{2m} \left[(i\vec{k}) \cdot (i\vec{k}) e^{i\vec{k} \cdot \vec{r}} - (i\vec{k})^2 e^{i\vec{k} \cdot \vec{r}} + V(\vec{r}) e^{i\vec{k} \cdot \vec{r}} \right] \right]$$

$$= \hat{H} + \frac{\hbar^2}{2m} \vec{k}^2 - \frac{i\hbar^2}{m} \vec{k} \cdot \nabla$$

(2) 对确定 n , $E_n(\vec{k})$ 是 $\vec{k}$ 的周期函数

$$E_n(\vec{k}) = E_n(\vec{k} + \vec{k}_n), \quad \psi_n(\vec{k}) = \psi_n(\vec{k} + \vec{k}_n)$$

$$\text{其中 } \vec{k}_n = \sum_i 2\pi b_i \vec{e}_i$$

$$\begin{aligned} \psi_{\vec{k}+\vec{k}_n}^n &= e^{i(\vec{k}+\vec{k}_n) \cdot \vec{r}} u_{\vec{k}+\vec{k}_n}^n(\vec{r}) \\ &= e^{i\vec{k} \cdot \vec{r}} \left[e^{i\vec{k}_n \cdot \vec{r}} u_{\vec{k}+\vec{k}_n}^n(\vec{r}) \right] \end{aligned}$$

$$\hat{H} \psi_{\vec{k}+\vec{k}_n}^n(\vec{r}) = E(\vec{k} + \vec{k}_n) \psi_{\vec{k}+\vec{k}_n}^n(\vec{r})$$

$$\hat{H}_n \left[e^{i\vec{k}_n \cdot \vec{r}} u_{\vec{k}+\vec{k}_n}^n(\vec{r}) \right] = E(\vec{k} + \vec{k}_n) \left[e^{i\vec{k}_n \cdot \vec{r}} u_{\vec{k}+\vec{k}_n}^n(\vec{r}) \right]$$

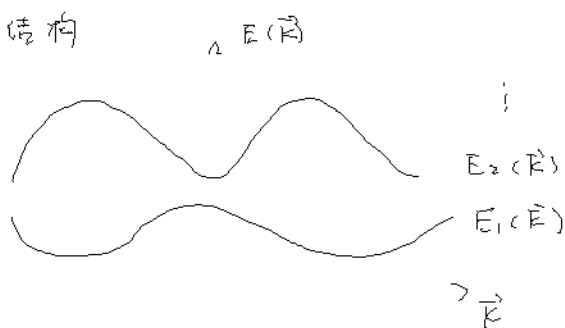
$$u_{\vec{k}+\vec{k}_n}^n = \left[e^{i\vec{k}_n \cdot \vec{r}} u_{\vec{k}+\vec{k}_n}^n(\vec{r}) \right]$$

$u_{\vec{k}+\vec{k}_n}^n$ 和 $u_{\vec{k}}^n$ 为相同本征值.

$$E(\vec{k} + \vec{k}_n) = E(\vec{k})$$

$$\rightarrow \vec{k} \in \text{BZ}$$

(3) 能谱成带结构



(4) 能谱对称性.

$$\hat{\sigma} E_n(\vec{k}) = E_n(\vec{k}), \quad \hat{\sigma} \text{ 为能谱的对称操作}$$

(5) 等能面垂直于布里渊空间.

§ 4.2 平面波法

单电子波函数: $\psi_{\vec{k}}(\vec{r}) = u_{\vec{k}}(\vec{r}) e^{i\vec{k} \cdot \vec{r}}$, $\vec{k} \in B$

$$u_{\vec{k}}(\vec{r} + \vec{R}_n) = u_{\vec{k}}(\vec{r})$$

按傅里叶展开,

$$u_{\vec{k}}(\vec{r}) = \frac{1}{\sqrt{\Omega}} \sum_n a_n(\vec{k}_n) e^{i\vec{k}_n \cdot \vec{r}}$$

$$\begin{aligned} \text{其中 } a_n(\vec{k}_n) &= \frac{1}{\sqrt{\Omega}} \int u_{\vec{k}}(\vec{r}) e^{-i\vec{k}_n \cdot \vec{r}} d\vec{r} \\ &= \frac{1}{\sqrt{\Omega}} \int \psi_{\vec{k}}(\vec{r}) e^{-i(\vec{k} + \vec{k}_n) \cdot \vec{r}} d\vec{r} \\ &= \frac{1}{\sqrt{\Omega}} \int \psi_{\vec{k} + \vec{k}_n}(\vec{r}) e^{-i(\vec{k} + \vec{k}_n) \cdot \vec{r}} d\vec{r} \\ &= a(\vec{k} + \vec{k}_n) \end{aligned}$$

$$\rightarrow \psi_{\vec{k}}(\vec{r}) = \frac{1}{\sqrt{\Omega}} \sum_n a(\vec{k} + \vec{k}_n) e^{-i(\vec{k} + \vec{k}_n) \cdot \vec{r}} d\vec{r}$$

$$|\psi_{\vec{k}}\rangle = \sum_n a_n(\vec{k} + \vec{k}_n) |\vec{k} + \vec{k}_n\rangle$$

$$\sum_n a_n(\vec{k} + \vec{k}_n) (T + V - E) |\vec{k} + \vec{k}_n\rangle = 0$$

$$V(\vec{r}) = \sum_n V(\vec{k}_n) e^{i\vec{k}_n \cdot \vec{r}} = \bar{V} + \sum_{n \neq 0} V(\vec{k}_n) e^{i\vec{k}_n \cdot \vec{r}}$$

$$= \bar{V} + \Delta V \approx \Delta V$$

$$\text{其中 } V(\vec{k}_n) = \frac{1}{\Omega} \int V(\vec{r}) e^{-i\vec{k}_n \cdot \vec{r}} d\vec{r}$$

$$\langle \vec{k} + \vec{k}_n | a(\vec{k} + \vec{k}_n) \left[-\frac{\hbar^2}{2m} \nabla^2 + V - E \right] | \vec{k} + \vec{k}_n \rangle = 0$$

$$\langle \vec{k} + \vec{k}_n | a(\vec{k} + \vec{k}_n) \left[\frac{\hbar^2}{2m} (\vec{k} + \vec{k}_n)^2 + V - E \right] | \vec{k} + \vec{k}_n \rangle = 0$$

$$\langle \vec{k} + \vec{k}_n | \vec{k} + \vec{k}_n \rangle = \delta_{\vec{k}_n, \vec{k}_n}$$

$$\sum_n \left\{ \left[\frac{\hbar^2}{2m} (\vec{k} + \vec{k}_n)^2 - E(\vec{k}) \right] \delta_{\vec{k}_n, \vec{k}_n} + \langle \vec{k} + \vec{k}_n | V | \vec{k} + \vec{k}_n \rangle \right\} a(\vec{k} + \vec{k}_n) = 0$$

$$\langle \vec{k} + \vec{k}_n | V | \vec{k} + \vec{k}_n \rangle = \frac{1}{\Omega} \int e^{-i(\vec{k} + \vec{k}_n) \cdot \vec{r}} V(\vec{r}) e^{i(\vec{k} + \vec{k}_n) \cdot \vec{r}} d\vec{r}$$

$$= V(\vec{k}_n - \vec{k}_n)$$

$$\left[\frac{\hbar^2}{2m} (\vec{k} + \vec{k}_n)^2 - E(\vec{k}) \right] a(\vec{k} + \vec{k}_n) + \underbrace{\sum_{n' \neq n}}_{\text{? 为代 } n' \neq n} V(\vec{k}_n - \vec{k}_{n'}) a(\vec{k} + \vec{k}_{n'}) = 0$$

$a(\vec{k} + \vec{k}_n)$ 有非平凡解

$$\det \left| \left[\frac{\hbar^2}{2m} (\vec{k} + \vec{k}_n)^2 - E(\vec{k}) \right] \delta_{\vec{k}_n, \vec{k}_{n'}} + V(\vec{k}_n - \vec{k}_{n'}) \right| = 0$$

$$\Rightarrow E_1(\vec{k}), \dots, E_n(\vec{k}) \dots$$

§ 4.3 近自由电子近似.

4.2 中的平面波法难以收敛, ΔV 为小量时, 电子近自由.

一、零级近似.

$$\left\{ \begin{aligned} \psi_{\vec{k}}^0(\vec{r}) &= \frac{1}{\sqrt{\Omega}} e^{i\vec{k} \cdot \vec{r}} \\ E^0(\vec{k}) &= \frac{\hbar^2}{2m} \vec{k}^2 \end{aligned} \right.$$

周期性边界条件: $\psi(\vec{r} + N_i \vec{a}_i) = \psi(\vec{r})$

$$\hookrightarrow N_i \vec{k} \cdot \vec{a}_i = 2\pi n_i$$

$$\vec{k} = \sum \frac{h_i}{N_i} \vec{b}_i$$

$$\text{状态密度} = \frac{1}{N_1} \cdot \left(\frac{\vec{b}_2}{N_2} \times \frac{\vec{b}_3}{N_3} \right) = \frac{N_1 N_2 N_3}{\Omega^*} = \frac{N\Omega}{(2\pi)^3}$$

$$\text{正交归一} \quad \left\{ \begin{aligned} \int \psi_{\vec{k}}^0(\vec{r}) \psi_{\vec{k}'}^0(\vec{r}) &= \delta_{\vec{k}, \vec{k}'} \\ \sum_{\vec{k}} \psi_{\vec{k}}^{0*}(\vec{r}) \psi_{\vec{k}}^0(\vec{r}') &= \delta(\vec{r} - \vec{r}') \end{aligned} \right.$$

二、简并微扰

$$\begin{aligned} \psi_{\vec{k}}(\vec{r}) &= \frac{1}{\sqrt{\Omega}} \sum_{\vec{k}'} a(\vec{k} + \vec{k}') e^{i(\vec{k} + \vec{k}') \cdot \vec{r}} \\ &= \frac{1}{\sqrt{\Omega}} \underbrace{a(\vec{k})}_{\approx 1} e^{i\vec{k} \cdot \vec{r}} + \sum_{\vec{k}' \neq 0} \frac{1}{\sqrt{\Omega}} \underbrace{a(\vec{k} + \vec{k}')}_{\text{小量}} e^{i(\vec{k} + \vec{k}') \cdot \vec{r}} \end{aligned}$$

$$\left[\frac{\hbar^2}{2m} (\vec{k} + \vec{k}')^2 - \underbrace{E(\vec{k})}_{\approx E^0(\vec{k}) = \frac{\hbar^2}{2m} \vec{k}^2} \right] a(\vec{k} + \vec{k}') + \sum_{\vec{k}' \neq 0} \underbrace{V(\vec{k}' - \vec{k})}_{\text{小量}} a(\vec{k} + \vec{k}') = 0$$

保留一阶小量:

$$a(\vec{k}) \approx 1$$

$$\left[\frac{\hbar^2}{2m} (\vec{k} + \vec{k}')^2 - \frac{\hbar^2}{2m} \vec{k}^2 \right] a(\vec{k} + \vec{k}') + V(\vec{k}') = 0$$

解得

$$V(\vec{k}')$$

$$a(\vec{k} + \vec{k}_n) = \frac{\hbar^2}{2m} \left[\vec{k}^2 - (\vec{k} + \vec{k}_n)^2 \right]$$

- 所近似的波函数为

$$\begin{aligned} \psi_{\vec{k}}(\vec{r}) &= \frac{1}{\Omega} e^{i\vec{k} \cdot \vec{r}} + \sum_{n \neq 0} \frac{1}{\Omega} V(\vec{k}_n) \frac{\hbar^2}{2m} \left[\vec{k}^2 - (\vec{k} + \vec{k}_n)^2 \right] e^{i(\vec{k} + \vec{k}_n) \cdot \vec{r}} \\ &= e^{i\vec{k} \cdot \vec{r}} u_{\vec{k}}(\vec{r}) \end{aligned}$$

$$\left[\frac{\hbar^2}{2m} (\vec{k} + \vec{k}_n)^2 - E(\vec{k}) \right] a(\vec{k} + \vec{k}_n) + \sum_{n'} \underbrace{V(\vec{k}_n - \vec{k}_{n'})}_{\text{小 } \frac{\hbar^2}{2m}} a(\vec{k} + \vec{k}_{n'}) = 0$$

= 所近似: $\vec{k}_n = 0$

$$V(-\vec{k}_n) = V(\vec{k}_n)$$

$$\left[\frac{\hbar^2}{2m} \vec{k}^2 - E(\vec{k}) \right] + \sum_{n \neq 0} \frac{\hbar^2}{2m} \left[\vec{k}^2 - (\vec{k} + \vec{k}_n)^2 \right] |V(\vec{k}_n)|^2 = 0$$

$$E^2(\vec{k}) = \frac{\hbar^2}{2m} \vec{k}^2 + \sum_{n \neq 0} \frac{\hbar^2}{2m} \left[\vec{k}^2 - (\vec{k} + \vec{k}_n)^2 \right] |V(\vec{k}_n)|^2$$

$$\text{其中 } V(\vec{k}_n) = \frac{1}{\Omega} \int V(\vec{r}) e^{-i\vec{k}_n \cdot \vec{r}} d\vec{r}$$

$$= \langle \vec{k} + \vec{k}_n | V | \vec{k} \rangle \quad \vec{k}, \vec{k} + \vec{k}_n \text{ 的散射矩阵元}$$

考虑周期势场 $V(\vec{r})$ 的扰动后,

电子的波函数 = 波矢为 $\vec{k}$ 的零级平面波 + $\vec{k} + \vec{k}_n$ 的散射波.

三、简并微扰 $\vec{k}^2 - (\vec{k} + \vec{k}_n)^2 = 0$ 非简并微扰发散.

$\vec{k}$ 态与 $\vec{k} + \vec{k}_n$ 存在能量上简并.

$$|\psi_{\vec{k}}\rangle = a(\vec{k}) |\vec{k}\rangle + a(\vec{k} + \vec{k}_n) |\vec{k} + \vec{k}_n\rangle$$

$$\left[\frac{\hbar^2}{2m} (\vec{k} + \vec{k}_n)^2 - E(\vec{k}) \right] a(\vec{k} + \vec{k}_n) + \sum_{n' \neq n} V(\vec{k}_n - \vec{k}_{n'}) a(\vec{k} + \vec{k}_{n'}) = 0$$

$$\left[\frac{\hbar^2}{2m} \vec{k}^2 - E(\vec{k}) \right] a(\vec{k}) + V(-\vec{k}_n) a(\vec{k} + \vec{k}_n) = 0 \quad k_n = 0$$

$$\left[\frac{\hbar^2}{2m} (\vec{k} + \vec{k}_n)^2 - E(\vec{k}) \right] a(\vec{k} + \vec{k}_n) + V(\vec{k}_n) a(\vec{k}) = 0 \quad k_n = k_n$$

为使有非零解：

$$\begin{vmatrix} \frac{\hbar^2}{2m} \vec{k}^2 - E(\vec{k}) & V(\vec{k}_n) \\ V(\vec{k}_n) & \frac{\hbar^2}{2m} (\vec{k} + \vec{k}_n)^2 - E(\vec{k}) \end{vmatrix} = 0$$

$$= \left[\frac{\hbar^2}{2m} \vec{k}^2 - E(\vec{k}) \right] \left[\frac{\hbar^2}{2m} (\vec{k} + \vec{k}_n)^2 - E(\vec{k}) \right] - \underbrace{V(\vec{k}_n) V(\vec{k}_n)}_{= V^2(\vec{k}_n)}$$

$$= E^2(\vec{k}) - \frac{\hbar^2}{2m} [\vec{k}^2 + (\vec{k} + \vec{k}_n)^2] E(\vec{k}) + \left(\frac{\hbar^2}{2m} \right)^2 \vec{k}^2 (\vec{k} + \vec{k}_n)^2 - 4 V^2(\vec{k}_n)$$

解得

$$\begin{cases} E_{\pm}(\vec{k}) = \frac{1}{2} \left\{ \frac{\hbar^2}{2m} [\vec{k}^2 + (\vec{k} + \vec{k}_n)^2] \pm \left[\left(\frac{\hbar^2}{2m} \right)^2 [\vec{k}^2 - (\vec{k} + \vec{k}_n)^2]^2 + 4 V^2(\vec{k}_n) \right]^{1/2} \right\} & k_n = k_n \\ E_{\pm}(\vec{k}) = \frac{\hbar^2}{2m} \vec{k}^2 \pm |V(\vec{k}_n)|^2 & k_n = 0 \end{cases}$$

四、布里渊区、能带、能隙、禁带

简并条件： $\vec{k}^2 = (\vec{k} + \vec{k}_n)^2$

$$\vec{k}_n \cdot (\vec{k} + \frac{1}{2} \vec{k}_n) = 0$$

在 E 空间中，解为 $-\vec{k}_n$ 的垂直平分面。

在 E_k 边界，有宽度为 $2 |V(\vec{k}_n)|$ 的能隙。



五. 能隙的形成

$$E'^2 = E^2, \quad E' = E + |U_1|$$

$$\text{波函数 } \varphi = \alpha \varphi_k^0(x) + \beta \varphi_{-k}^0(x)$$

$$\begin{cases} (E_k^0 - E) \alpha + U_1 \beta = 0 \\ U_1 \alpha + (E_{-k}^0 - E) \beta = 0 \end{cases}$$

$$\begin{vmatrix} E_k^0 - E & U_1 \\ U_1 & E_{-k}^0 - E \end{vmatrix} = 0 \quad \text{解得 } E_{\pm} = E_k^0 \pm |U_1|$$

$$\text{代回: } \left(\frac{\alpha}{\beta}\right)_{\pm} = \pm \frac{U_1}{|U_1|} = \mp 1 \quad (U_1 \text{ 为实数, } U_1 < 0)$$

$$\varphi_+(x) = \alpha [\varphi_k^0(x) - \varphi_{-k}^0(x)] = \alpha (e^{-i\lambda x/a} - e^{i\lambda x/a}) = \sqrt{\frac{2}{L}} \sin\left(\frac{\lambda x}{a}\right)$$

$$\varphi_-(x) = \sqrt{\frac{2}{L}} \cos\left(\frac{\lambda x}{a}\right), \quad k = -\frac{\lambda}{a}, \quad k' = \frac{\lambda}{a}$$

相应电子概率分布:

$$\rho_+(x) = \frac{2}{L} \sin^2\left(\frac{\lambda x}{a}\right)$$

$$\rho_-(x) = \frac{2}{L} \cos^2\left(\frac{\lambda x}{a}\right)$$

$$\rho(x) \propto \rho_+ \quad \rho_-$$



两种驻波中电子倾向于聚集位置不同.

$$\text{取 } U(x) = 2U_1 \cos\left(\frac{2\lambda x}{a}\right)$$

$$\text{能隙宽度 } E_g = \int U(x) (\rho_+ - \rho_-) dx$$

$$= \int_0^L 2U_1 \cos\left(\frac{2\lambda x}{a}\right) \frac{2}{L} \left[\sin^2\left(\frac{\lambda x}{a}\right) - \cos^2\left(\frac{\lambda x}{a}\right) \right] dx$$

$$=$$

$$= -2U_1 = 2|U_1|$$

在 3D 晶格中.

$$V(\vec{r}) = \sum_{\vec{R}_1} \sum_{\vec{R}_2} U(\vec{r} - \vec{R}_1 - \vec{R}_2)$$

$$V(\vec{r}) = V(\vec{R}_1 + \vec{r})$$

$$V(\vec{k}) = \frac{1}{N\Omega} \int \sum_{\vec{R}_1} \sum_{\vec{R}_2} U(\vec{r} - \vec{R}_1 - \vec{R}_2) e^{-i\vec{k} \cdot \vec{r}} d\vec{r}$$

$$= \frac{1}{N\Omega} \sum_{\vec{R}_1} \int U(\vec{r}) e^{-i\vec{k} \cdot (\vec{R}_1 + \vec{r} + \vec{r})} d\vec{r}$$

$$= \frac{1}{N\Omega} \sum_{\vec{R}_1} e^{-i\vec{k} \cdot (\vec{R}_1 + \vec{r})} \int U(\vec{r}) e^{-i\vec{k} \cdot \vec{r}} d\vec{r}$$